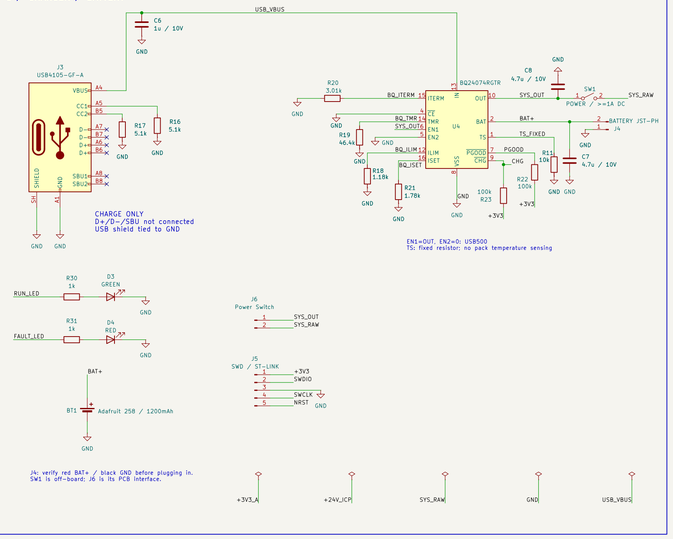
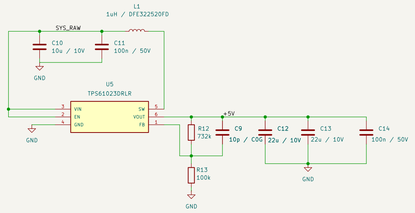
USB-C / CHARGER / BATTERY



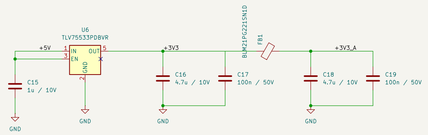
J4: verify red BAT+ / black GND before plugging in.
SW1 is off-board; J6 is its PCB interface.

5 V BOOST



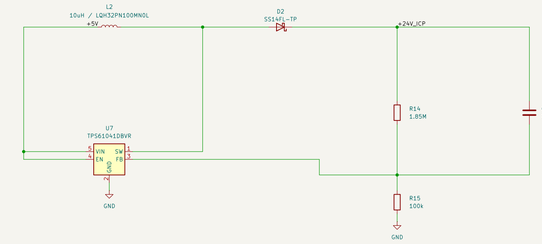
VOUT = 0.6 × (1 + 732k/100k) = 4.992 V

3.3 V / ANALOG RAIL



AGND region uses the same GND net.
One continuous PCB ground plane.

24 V ICP BOOST



VOUT = 1.233 × (1 + 1.85M/100k) = 24.04 V

ICP ACCEL / AC COUPLING / BUFFER

SWD pin 1 = VTREF (target voltage sense).
Buttons active-low: LEDs active-high. SWO is a spare test node.

Q1 pins 2 + 3 joined. No low-voltage TVS on ICP.
Do not place BAT54S on the 1Mohm-biased input.

ADS131M02 / INPUT FILTER / CLOCK

AGND pin 2 and DGND pin 19 share GND.
PGA=8 initial setting; measure and compensate filter delay.

STM32G491CET6 / MCU DECOUPLING

C decouplers: VDD24, VDD36, VDD48, VBAT1.
Pin 21 VDDA + pin 20 VREF+ use +3V3_A.

HSE BYPASS / PF1 NC
PA0 = TIM2_CH1 / AF1
SWD enabled: PB8 boot pulldown

OPTICAL TACH / FILTERED 5 V

PHOTO TACH: PIN A +5V — VERIFY CONNECTOR ORIENTATION

DS5 1097: A=+5V, B=GND, D=signal.
Resettable fuse is 100mA hold; trip current is higher.

DISPLAY + ON-MODULE microSD

DISPLAY VIN = +3V3_DISP from the U9 TPS62260 buck (bottom right).
Module datasheet: 3.3 V VIN; 5 V must not be applied.

4-wire SPI: close module R2/R3/R5; open R1/R4/R6.
LCD_DC = J1.3 WRX. J1.2 is the clock alias, left open.
Touch and card-detect not used. Separate SD socket removed.

DISPLAY 3.3 V BUCK

Feeds DS1 J2.2 VIN (+3V3_DISP). Module VDD 3.0–3.6 V, 200 mA typ / 280 mA max; 5 V must not be applied.
TPS62260 2.25 MHz, 600 mA, EN tied to VIN. L3 DFE322520FD-2R2M; C45 22 pF feed-forward across R32.

VOUT = 0.6 × (1 + 453k/100k) = 3.32 V

Display buck U9 added 2026-09-08: bench validation pending before fabrication

Sheet: /
File: Balancer.kicad_sch

Title: Balancer Rev A — wired prototype

Size: A1	Date:	Rev: A / REVIEW
KiCad E.D.A. 10.0.3		Id: 1/1